PARISHRAM 2.0 2026 (Class 12th)

CHEMISTRY

DPP 03

d & f-Block Elements

Q1 The highest magnetic moment is shown by the transition metal ion with the outer electronic configuration
(A) $3d^2$ (B) $3d^7$
(C) $3d^5$ (D) $3d^9$

Q2 Electrons in a paramagnetic compound are
(A) unpaired (B) shared
(C) paired (D) donated

Q3 Among the following outermost configurations of transition metals, which shows the highest oxidation state?
(A) $3d^5 4s^1$ (B) $3d^3 4s^2$
(C) $3d^5 4s^2$ (D) $3d^6 4s^2$

Q4 The coloured ions are
(A) Sc^{3+}, Cu^{2+}
(B) V^{3+}, Ni^{3+}
(C) Zn^{2+}, Mn^{7+}
(D) Cd^{2+}, Cr^{3+}

Q5 Which of the following ions is coloured in solution?
(A) Zn^{2+} (B) Ti^{4+}
(C) Cu^{+} (D) V^{2+}

Q6 Which one of the following metals is used as a catalyst in the Haber's process?
(A) Tungsten
(B) Molybdenum
(C) Chromium
(D) Iron containing Molybdenum

Q7 The property, which is not characteristic of transition metals is
(A) variable oxidation states
(B) tendency to form complexes
(C) formation of coloured compounds
(D) natural radioactivity

Q8 German silver has constituents
(A) Ag,Ni,Zn (B) Cu,Zn,Sn
(C) Cu,Zn,Ni (D) Fe,Zn,Ni

Q9 Cu, Ag and Au are
(A) inner transition elements
(B) elements with oxidation states +3
(C) non-transition elements
(D) coinage metals

Answer Key

Q1	C
Q2	A
Q3	C
Q4	B
Q5	D
Q6	D
Q7	D
Q8	C
Q9	D

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

$3d^5$ has maximum no. of unpaired electrons.

Video Solution:

Q2 Text Solution:

Electrons in a paramagnetic compound are unpaired.

Video Solution:

Q3 Text Solution:

In the first transition series Mn[$3d^5$ $4s^2$] shows the maximum oxidation state of 7.

Video Solution:

Q4 Text Solution:

V^{3+}, Ni^{3+}

Video Solution:

Q5 Text Solution:

V^{2+} ions is coloured in solution.

Video Solution:

Q6 Text Solution:

In the Haber's process, which is used for the industrial production of ammonia (NH_3), the catalyst typically used is iron containing small amounts of promoters such as molybdenum (Fe-Mo catalyst). This catalyst is crucial in facilitating the reaction between nitrogen (N_2) and hydrogen (H_2) to form ammonia (NH_3).

Video Solution:

Q7 Text Solution:

The property, which is not characteristic of transition metals is natural radioactivity.

Video Solution:

Q8 Text Solution:

German silver has constituents Cu,Zn,Ni.

Video Solution:

Q9 **Text Solution:**
coinage metals

Video Solution:

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